

WASP-104b

R-2129

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_250309 · created 2026-08-11T23:16:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460743.71 +/- 0.0063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.087 +/- 0.035
Transit depth (Rp/Rs)^2	0.75 +/- 0.6 [%]
Orbital Inclination (inc)	87.21 +/- 2.99
Airmass coefficient 1 (a1)	1.21 +/- 0.017
Airmass coefficient 2 (a2)	-0.1422 +/- 0.0099
Scatter in the residuals of the lightcurve fit is	1.45 %
Best Comparison Star	None
Optimal Aperture	3.76
Optimal Annulus	4.98
Transit Duration (day)	0.084 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.087 ± 0.035 vs 0.1214 (deviation 0.98σ)
Duration fit vs published	0.084 d vs 0.0742 d (deviation 0.66σ)
Mid-time O–C	-13.7 min
Depth SNR	2.5
Residual scatter	1.45 %

Light curve

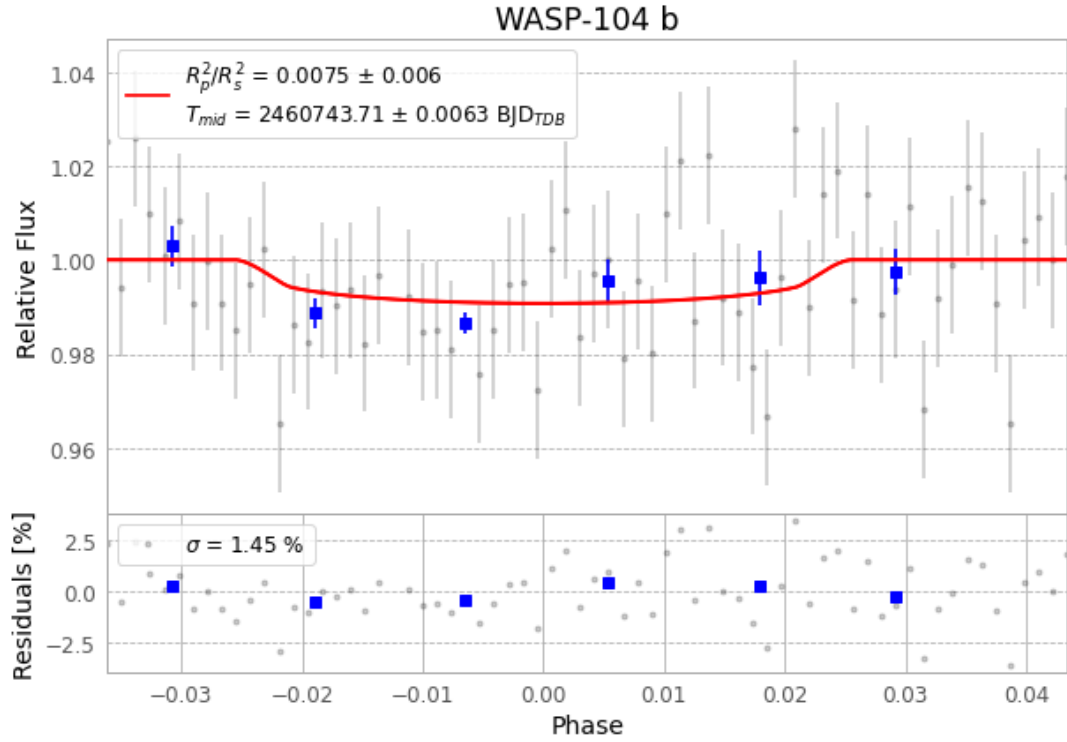


Figure 1 — FinalLightCurve_WASP-104 b_2025-03-08.png (SHA-256 94deac84933fae73...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 215], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 9624788a89bea546...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

68 FITS frames (45 MB), WASP-104250309032115.FITS → WASP-104250309064215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-250309.log (bb02c9f35753...), FinalLightCurve_WASP-104 b_2025-03-08.png (94deac84933f...), FinalLightCurve_WASP-104 b_2025-03-08.csv (9ccf25482d97...)

Compute resources

Wall time 187.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 23:16Z.